

Open

Decouple gRPC from the worker protocol and add an in-process transport #573

stuhood wants to merge 4 commits into

[datafusion-contrib:main](#) from [paradedb:stuhood.in-memory-trans...](#) 

View status

Code ▾

Decouple gRPC from the worker protocol and add an in-process transport #573

Open

stuhood wants to merge 4 commits into

[datafusion-contrib:main](#) from [paradedb:stuhood.in-memory-trans...](#)  Conversation 11 Commits 4 Checks 0 Files changed**stuhood** commented 2 days ago • edited ▾

What

Decouples the `datafusion-distributed` core logic from the `grpc` transport and introduces an in-process transport for single-process distributed task execution.

Why

Previously, the worker protocol and the `grpc` feature were coupled, which meant any use of the distributed execution abstraction required the `tonic` gRPC stack. Separating the transport layer from the protocol messages allows the framework to be used with alternative transports. The in-process transport provides a mechanism to run and test the abstraction when the `grpc` feature is disabled.

How

- Split `worker.proto` into `messages.proto` (data structures) and `worker_service.proto` (gRPC service definition in the `worker_grpc` package). This splits the generated code into `worker.rs` (always compiled) and `worker_grpc.rs` (compiled only when the `grpc` feature is enabled).
- Added `InProcessChannelResolver`. It implements `WorkerChannel` by routing the three protocol methods directly to a co-located `Worker` instance in the same process, bypassing IPC, networking, and serialization.
- Build adjustments:
 - Removed the `benchmarks` dev-dependency to prevent `grpc` from being enabled in every test build.
 - Gated gRPC-coupled test utilities in `test_utils` behind the `grpc` feature.

Open

Decouple gRPC from the worker protocol and add an in-process transport #573

stuhood wants to merge 4 commits into

datafusion-contrib:main from paradedb:stuhood.in-memory-trans... 

- Added a `unit-test-no-grpc` CI job that verifies the core library and the in-process transport compile and pass unit tests without the `grpc` feature.
- Added an end-to-end test for `InProcessChannelResolver` (a distributed `GROUP BY` across tasks) that runs in the no-gRPC CI job.

 **mdashti** and others added 4 commits 2 days ago

-   Moved the worker-protocol messages out from behind the `grpc` feature. ... 4721f09
-   Made the no-gRPC build compile and its unit suite run in CI. ... b0a7e4a
-   Added an in-process worker transport that needs no gRPC. ... 5f41597
-   Split `worker.proto` into `messages.proto` and `worker_service.proto`. 3888470

  **stuhood** marked this pull request as ready for review 2 days ago  **stuhood** force-pushed the `stuhood.in-memory-transport` branch from **9be56b0** to **3888470** 2 days ago [Compare](#)  **stuhood** mentioned this pull request 2 days ago

feat: add an in-process worker transport (no gRPC) #548

 Closed **gabotechs** reviewed yesterday[View reviewed changes](#)**gabotechs** left a comment

Collaborator

Open

Decouple gRPC from the worker protocol and add an in-process transport #573

stuhood wants to merge 4 commits into

datafusion-contrib:main from paradedb:stuhood.in-memory-trans... 

Ship incrementally. Identify the following pieces of work in the following order.

1. split worker.proto into worker_service.proto and messages.proto , putting some thought into where to place the proto generator script so that we don't copy-paste too much.
2. Ship the InProcessWorkerChannel , and wire it up to channel_resolver.rs (or somewhere else that makes sense) so that we can start executing queries without grpc . Probably, the interaction of InProcessWorkerChannel and LocalWorkerContext needs to happen here.
3. Tweak existing tests unit tests that rely on grpc so that they just use in-process comms (mainly the tests about rewriting plans with metrics). That will allow to decouple them for gRPC by using the in-process implementation.
4. Tweak the remaining test utils, moving some to grpc/ and flagging them appropriately:
 - If grpc feature is enabled, then gRPC worker channels are used (still with in-process fallbacks for self urls)
 - If grpc is disabled, exclusively in-process worker channels are used. This should be transparent to the tests.



▼ console/src/main.rs

Comment on lines +66 to 70

	66	+	if event::poll(Duration::from_millis(16))?
	67	+	<u>&&</u> let Event::Key(key) = event::read()?
	68	+	{
	69	+	<u>input::handle_key_event(app, key);</u>
70	70		}



gabotechs 2 days ago

Collaborator

🤔 this change is unrelated right? if it is, I'd avoid introducing unrelated changes, mainly for better traceability.



Reply...

Resolve conversation

Open

Decouple gRPC from the worker protocol and add an in-process transport #573

stuhood wants to merge 4 commits into

datafusion-contrib:main from paradedb:stuhood.in-memory-trans... 

```
179 | +           if old_task.status == taskStatus::Running as 132
180 | +               && let Some(sk) = &old_task.task_key
181 | +           {
```



gabotechs 2 days ago

Collaborator

Same as above



Reply...

Resolve conversation

✓ src/protocol/grpc/gen/src/main.rs

```
8 | - let proto_file = proto_dir.join("worker.proto");
9 | - let out_dir = repo_root.join("src/protocol/grpc/generated");
  | 7 | + let protocol_dir = repo_root.join("src/protocol");
  | 8 | + let messages_proto = protocol_dir.join("messages.proto");
```



gabotechs 2 days ago

Collaborator

This generator file is under the `grpc` scope, but it's in charge of generating `messages.proto` which is not part of `grpc`.

Probably copy-pasting this generator script one folder above is not worth it. What about moving it to the root of the project under a `codegen/` folder or something like that? we can centralize proto generation there.



Reply...

Resolve conversation

✓ src/protocol/generated/mod.rs

```
| 2 | +
```

Open

Decouple gRPC from the worker protocol and add an in-process transport #573

stuhood wants to merge 4 commits into

[datafusion-contrib:main](#) from [paradedb:stuhood.in-memory-trans...](#) 

gabotechs 2 days ago

Collaborator

If this is specific to `grpc`, ideally it should be generated under the `grpc/` folder



Reply...

Resolve conversation

▼ `src/test_utils/in_memory_channel_resolver.rs`

Comment on lines +29 to +31

```
29 + #[cfg(feature = "grpc")]
30 + mod channel {
31 +     use super::InMemoryWorkerResolver;
```



gabotechs 2 days ago

Collaborator

All the test utils that are specific to `grpc` should probably live under the `grpc/` folder, that way they are gated automatically.



Reply...

Resolve conversation

▼ `src/worker/mod.rs`

Comment on lines +6 to 9

```
6 + // `worker_handles` builds `tonic` servers and Flight channels for the be
7 + // only compile with the gRPC transport.
8 + #[cfg(all(feature = "grpc", any(test, feature = "integration")))]
7 9 pub(crate) mod test_utils;
```



gabotechs 2 days ago

Collaborator

Open

Decouple gRPC from the worker protocol and add an in-process transport #573

stuhood wants to merge 4 commits into

datafusion-contrib:main from paradedb:stuhood.in-memory-trans... 

Reply...

Resolve conversation

src/coordinator/metrics_store.rs

```
28 |     28 |     }  
29 |     29 |  
30 |     -   #[cfg(test)]  
    +   30 | +   #[cfg(all(test, feature = "grpc"))]
```



gabotechs yesterday

Collaborator

Seems like this is needed because this code is used in the task_metrics_rewriter.rs tests, and those tests are still coupled to grpc .

It'd be nice to decouple that test from grpc first so that we can avoid flagging this here.



Reply...

Resolve conversation

benchmarks/tests/clickbench_correctness_test.rs

```
... | ... | @@ -1,4 +1,4 @@  
1 | -   #[cfg(all(feature = "integration", feature = "clickbench", test))]  
  | +   1 | +   #[cfg(all(feature = "clickbench", test))]
```



gabotechs yesterday

Collaborator

These tests should really not be under benchmarks. I think I might be missing the reason for these to be here. Do they really need to be here?



Open

Decouple gRPC from the worker protocol and add an in-process transport #573

stuhood wants to merge 4 commits into

datafusion-contrib:main from paradedb:stuhood.in-memory-trans... 

src/execution_plans/metrics.rs

```
23 | 23 |  
24 | -   #[cfg(test)]  
    | +   #[cfg(all(test, feature = "grpc"))]  
25 | 25 | pub(crate) fn inner(&self) -> &Arc<dyn ExecutionPlan> {
```



gabotechs yesterday

Collaborator

This is only used in task_metrics_rewriter.rs tests. If we manage to decouple those tests from grpc, we can leave this untouched.



Reply...

Resolve conversation

src/protocol/in_process.rs

```
30 | + /// state for every task, keyed by [`crate::TaskKey`], the same way the g  
31 | + /// tasks of a query land on it.  
32 | + #[derive(Clone)]  
33 | + pub struct InProcessChannelResolver {
```



gabotechs yesterday

Collaborator

For a follow-up PR: this should somehow interact with channel_resolver.rs and LocalWorkerContext so that, once a request is made to a URL that matches LocalWorkerContext.self_url, the InProcessChannelResolver kicks in automatically.



Reply...

Resolve conversation

Open

Decouple gRPC from the worker protocol and add an in-process transport #573

stuhood wants to merge 4 commits into

[datafusion-contrib:main](#) from [paradedb:stuhood.in-memory-trans...](#) 



Review required

At least 1 approving review is required by reviewers with write access.

1 workflow awaiting approval

This workflow requires approval from a maintainer. [Learn more about approving workflows.](#)

Merging is blocked

At least 1 approving review is required by reviewers with write access.

Reviewers



[gabotechs](#)

Still in progress? [Convert to draft](#) 

At least 1 approving review is required to merge this pull request.

Still in progress? [Convert to draft](#)

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Open**Decouple gRPC from the worker protocol and add an in-process transport #573**

stuhood wants to merge 4 commits into

[datafusion-contrib:main](#) from [paradedb:stuhood.in-memory-trans...](#) 

3 participants